"I think the range of manoeuvres they can do is by far the largest" in the autonomous helicopter field, said Eric Feron, a Georgia Tech aeronautics and astronautics professor who worked on autonomous helicopters while at MIT. "But what's more impressive is the technology that underlies this work. In a way, the machine teaches itself how to do this by watching an expert pilot fly. This is amazing."

Writing software for robotic helicopters is a daunting task, in part because the craft itself, unlike an airplane, is inherently unstable. "The helicopter doesn't want to fly. It always wants to just tip over and crash," said Oku, the pilot.

To scientists, a helicopter in flight is an "unstable system" that comes unglued without constant input. Abbeel compares flying a helicopter to balancing a long pole in the palm of your hand: "If you don't provide feedback, it will crash."

Early on in their research, Abbeel and Coates attempted to write computer code that would specify the commands for the desired trajectory of a helicopter flying a specific manoeuvre. While this hand-coded approach succeeded with novice-level flips and rolls, it flopped with the complex tic-toc.

It might seem that an autonomous helicopter could fly stunts by simply replaying the exact finger movements of an expert pilot using the joy sticks on the helicopter's remote controller. That approach, however, is doomed to fail because of uncontrollable variables such as gusting winds.

When the Stanford researchers decided that their autonomous helicopter should be capable of flying airshow stunts, they realized that even defining their goal was difficult. What's the formal specification for "flying well?" The answer, it turned out, was that "flying well" is whatever an expert radio control pilot does at an airshow.

So the researchers had Oku and other pilots fly entire airshow routines while every movement of the helicopter was recorded. As Oku repeated a manoeuvre several times, the trajectory of the helicopter inevitably varied slightly with each flight. But the learning algorithms created by Ng's team were able to discern the ideal trajectory the pilot was seeking. Thus the autonomous helicopter learned to fly the routine better – and more consistently – than Oku himself.

During a flight, some of the necessary instrumentation is mounted on the helicopter, some on the ground. Together, they continuously